

Cross Asset Research Suite

User tutorial: step-by-step setup, interpretation, and practical workflows

Guide for the fixed Pine v6 TradingView indicator

Version note: aligns with the fixed Cross Asset Research Suite script and current TradingView documentation available in April 2026.

At a glance	Summary
What it teaches	How to load the indicator, choose a pair, configure the inputs, interpret each view, and create alerts.
Who it is for	Traders and researchers who want a repeatable cross-asset workflow without leaving TradingView.
Fastest path	Choose a linked pair, start on the chart timeframe you care about, pick a spread model, then read correlation before acting on z-score or leadership.
Most common mistake	Using a weak or unstable pair and then over-trusting the z-score.

This tutorial is intentionally operational. It assumes the indicator is already installed in TradingView and focuses on how to use it correctly, how to read it, and how to avoid the most common interpretation errors.

Quick-start workflow

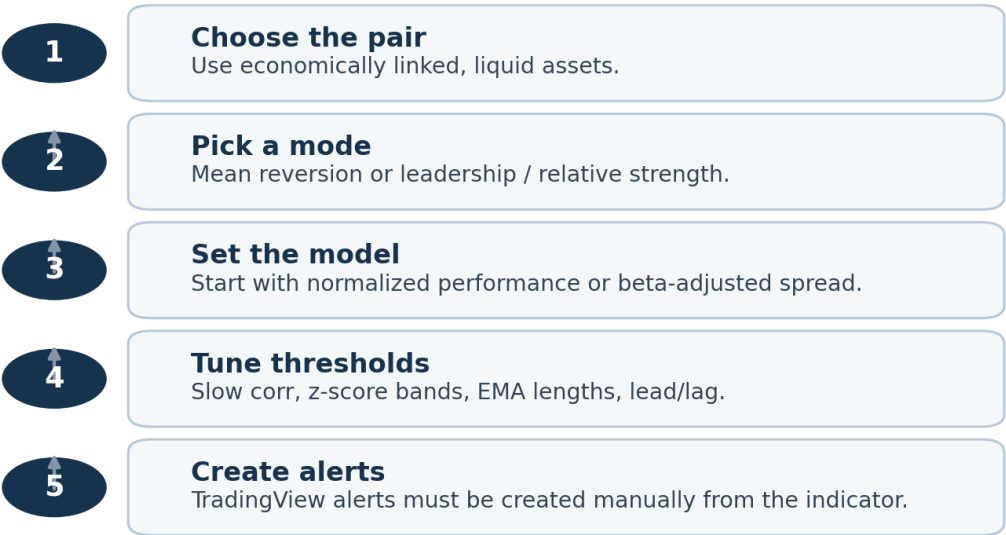


Figure 1. Five-step quick-start workflow.

1. Before you begin

- Use liquid instruments first. Thin symbols create noisy ratios and fragile correlations.
- Pick assets with a real economic link. Good examples are stock vs benchmark, sector ETF vs broad market, or two highly related instruments within the same theme.
- Set the chart to the timeframe you actually want to analyze. XARS uses the chart timeframe for its core engine by design, which matches TradingView's `request.security()` guidance.

Why this matters: TradingView recommends `request.security()` primarily for equal or higher timeframes, while lower-timeframe work should use `request.security_lower_tf()`. Since the indicator is built around chart-timeframe alignment, the easiest way to control the analysis horizon is simply to change the chart timeframe.^[2]

Most important habit: If the pair looks wrong on the chart, do not fix that with more settings. First fix the pair choice, the session context, or the timeframe.

2. What each input group does

Input group	What you control	Practical note
Symbols	Asset A, Asset B, whether to use the chart symbol as A, and source mode.	This is where you define the pair and the price source that feeds it.
Cross-analysis engine	Spread model, normalization lookback, correlation lengths, beta length, z-score length, ratio EMAs, RS momentum, and lead/lag parameters.	These settings control the math and the sensitivity.
Signals	Bar-close confirmation, mean-reversion enable/disable, leadership enable/disable, z-score thresholds, minimum slow correlation, and chart markers.	These settings control what qualifies as a signal.
Display	Pane view, regime shading, and dashboard visibility.	Use these to switch between diagnostic modes without changing the underlying data.

3. Step-by-step setup

Step 1 — Choose Asset A and Asset B

Asset A is the instrument you want to evaluate; Asset B is the comparator or hedge leg. If 'Use chart symbol as Asset A' is on, the chart symbol becomes Asset A automatically. This is convenient when you want to drop the indicator on different charts and keep the same comparator.

- Stock vs benchmark: chart = NVDA, Asset B = QQQ.
- Sector vs market: chart = XLE, Asset B = SPY.
- Relative strength within a theme: chart = SMH, Asset B = QQQ.

Step 2 — Pick a source mode

The script lets you use Close, HL2, HLC3, or OHLC4. Start with Close unless you have a specific reason not to. Averaged price sources can smooth some noise, but they also make the indicator slightly less comparable to decisions based on the actual close.

Step 3 — Choose the spread model

Model	When to start with it	How to think about it
Normalized performance difference	Best first choice for classic pair divergence.	Both assets are indexed to 100 over the lookback, then compared as a distance.
Price ratio	Best first choice for leadership or simple relative strength.	Higher values mean Asset A is outperforming Asset B in raw ratio terms.
Log ratio	Use when the pair has very different absolute price levels.	A cleaner transformation of the raw ratio.
Beta-adjusted log spread	Best first choice for a more hedge-aware pair spread.	Attempts to account for how strongly A tends to move relative to B.

Step 4 — Set the core lookbacks

Use slow settings before you use aggressive settings. The goal is to stabilize the reading first. A good baseline is 60 bars for slow correlation, beta, and z-score, with 20 bars for fast correlation and 10/30 for the ratio EMAs.

Step 5 — Decide whether you are trading reversion or leadership

This is the fork in the road. If you are looking for a divergence that may normalize, focus on slow correlation, the z-score bands, and the MR alerts. If you are looking for the stronger leg in a pair, focus on the price ratio, fast vs slow ratio EMA, RS momentum, and the leadership alerts.

4. How to read each pane view

4.1 Z-score view

Use this view for mean reversion. The orange line is the standardized distance of the chosen spread model from its rolling mean. The upper and lower entry lines are the mean-reversion trigger bands, and the inner gray lines represent the exit zone.

- Above the positive entry band: the spread is unusually high relative to its recent history.
- Below the negative entry band: the spread is unusually low relative to its recent history.
- Back inside the exit band: the move has substantially normalized.

4.2 Normalized performance view

Use this view to see which asset has performed better from a common starting point. Both lines are rebased to 100 over the normalization lookback. If the lines separate, the pair has diverged in

normalized terms. This is the most intuitive view for discretionary pair comparison and the closest to the classical distance framework.^[10]

4.3 Correlation view

Use this view to answer one question first: does the pair still deserve a pair-trading interpretation? TradingView's correlation coefficient scale runs from -1 to +1. In practice, positive and stable slow correlation makes a mean-reversion read more credible, while a breakdown below your minimum threshold is a regime warning.^[7]

4.4 Leadership view

Use this view to answer a different question: which asset is leading right now? Here the indicator shows the EMA spread percentage of the ratio, the ratio momentum, and the lead/lag score. Positive readings favor Asset A leadership; negative readings favor Asset B leadership.

4.5 Raw spread/ratio view

Use this when you want to inspect the chosen core series directly without standardization. It is especially useful for visually checking whether the moving mean in the selected spread model is stable or drifting.

5. How to read the dashboard and chart markers

Field	What it means	How to use it
Fast corr	Short-horizon co-movement.	Use as a temperature check, not the main gate.
Slow corr	Primary relationship filter.	If this is weak, pair-trading signals are less trustworthy.
Beta	Rolling sensitivity of A returns to B returns.	Most important when using the beta-adjusted spread.
Z-score	Standardized stretch of the chosen spread model.	Main input for mean-reversion decisions.
RS momentum	Momentum of the price ratio.	Helpful for confirming leadership.
Lead/Lag	Proxy for whether one asset's returns tend to precede the other's.	Exploratory; use as supporting evidence only.
Regime	Script's summary of whether the pair looks MR-ready, trend-like, or unstable.	Read this before acting on a signal.
Bias	Current directional summary: long A/short B, long B/short A, A leading, B leading, or neutral.	A concise final check.

Main-chart markers are plotted with `force_overlay` so they can appear on price even though the analytics run in a separate pane. That is why the program can remain readable without sacrificing visibility on the chart itself.^[4]

6. Practical workflows

Workflow A — Mean reversion on a stock vs benchmark

1. Open the stock chart you care about, such as AAPL or NVDA.
2. Set Asset B to QQQ or SPY.
3. Start with Beta-adjusted log spread.
4. Use 60 for slow correlation, beta, and z-score; 20 for fast correlation.
5. Require slow correlation above the minimum threshold before trusting the z-score.
6. Treat a z-score stretch beyond ± 2.0 as attention, not automatic execution.
7. Look for the exit signal only after the z-score compresses meaningfully.

Workflow B — Relative-strength leadership between sector and market

1. Use a sector ETF on the chart, such as XLF, XLK, or XLE.
2. Set Asset B to SPY.
3. Switch the spread model to Price ratio or Log ratio.
4. Pay more attention to ratioFast vs ratioSlow and RS momentum than to the z-score.
5. Use trendA or trendB markers to identify leadership crossovers.

Workflow C — Research mode on daily charts

1. Move the chart to 1D.
2. Use Normalized performance difference first to visually inspect long-horizon tracking.
3. Then switch to Beta-adjusted log spread to see whether the divergence remains meaningful after beta adjustment.
4. Use the dashboard to decide whether the pair is strong enough for a deeper external backtest.

7. Setting up alerts the right way

The script exposes `alertcondition()` triggers, but Pine does not create alerts for you. In TradingView, you still need to open the Create Alert dialog, choose the script in the Condition dropdown, and then select the specific trigger you want. If you want several triggers, create them separately.^[3]

1. Add the indicator to the chart.
2. Open TradingView's Create Alert dialog.
3. Choose Cross Asset Research Suite in the Condition field.
4. Pick the desired trigger, such as 'MR Long A / Short B' or 'A leadership cross'.
5. Choose your alert frequency and expiration.
6. Repeat for additional triggers if needed.

Troubleshooting: If an alert did not fire, check three things first: the alert still exists, the chosen trigger was correct, and the bar-close confirmation setting matches your expectation.

8. Tuning guide

If you see this	Likely cause	What to change first
Too many MR signals	Thresholds are too loose or the pair is noisy.	Raise meanRevEntryZ, increase zLen, or choose a stronger pair.
Good-looking z-score but poor outcomes	Correlation regime is weak or unstable.	Raise minCorrForMR and check the pair choice.
Leadership signals feel late	Trend filters are too slow for the timeframe.	Reduce ratioFastLen or ratioSlowLen slightly.
Signals flip too often	The timeframe is too noisy or confirmation is off.	Use bar-close confirmation and/or move up a timeframe.
Dashboard looks stale or overloaded	You are reading too many modes at once.	Focus on one pane view and one regime at a time.

9. Common mistakes

- Using unrelated assets just because their recent chart paths look similar.
- Treating a high correlation as proof of a robust spread.
- Changing every setting before deciding whether the pair itself is worth studying.
- Mixing mean-reversion logic and leadership logic in the same trade idea.
- Ignoring event risk, earnings, or contract roll effects.

10. Daily checklist

1. Pick a pair with a real economic link.
2. Set the chart timeframe to the horizon you care about.
3. Choose the spread model that matches your objective.
4. Check slow correlation before trusting any MR signal.
5. Read the dashboard bias and regime text.
6. Create only the alerts you actually plan to act on.
7. Review signals in the context of the main chart, not in isolation.

11. Final takeaway

The fastest way to become consistent with Cross Asset Research Suite is to keep the workflow simple: first choose a good pair, then decide whether the job is mean reversion or leadership, and only then adjust the settings. When the pair is sound and the mode is clear, the indicator becomes much easier to trust.

References

Accessed April 2026.

- [1] [TradingView User Manual — Other timeframes and data](#) — Dynamic requests in Pine v6, `request.security()`, and related `request.*()` behavior.
- [2] [TradingView FAQ — Other data and timeframes](#) — Lower-timeframe limitations, `request.security_lower_tf()`, and the recommended higher-timeframe non-repainting pattern.
- [3] [TradingView FAQ — Alerts](#) — `alertcondition()` behavior and the need for users to create alerts from the Create Alert dialog.
- [4] [TradingView User Manual — Visuals overview](#) — `overlay` and `force_overlay` behavior for mixed-pane designs.
- [5] [TradingView User Manual — Tables](#) — Best practice for declaring tables with `var` and updating them on the last bar.
- [6] [TradingView User Manual — Limitations](#) — `request.*()` call limits, tuple element limits, and `calc_bars_count` behavior.
- [7] [TradingView Help Center — Correlation Coefficient \(CC\)](#) — Meaning of the correlation scale from -1 to +1 in chart analysis.
- [8] [TradingView User Manual — Profiling and optimization](#) — Guidance on minimizing `request.*()` calls and using tuple requests for efficiency.
- [9] [TradingView User Manual — Inputs](#) — `input.symbol()`, `input.*` grouping, and user-facing configuration design.
- [10] [Gatev, Goetzmann, and Rouwenhorst \(2006\), Pairs Trading: Performance of a Relative-Value Arbitrage Rule](#) — Classic reference for distance-based pair selection using normalized historical prices.